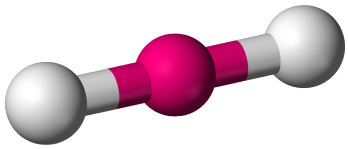
7
Chemical bond



IN nature molecules can found in a myriad of different shapes. Some like carbon dioxide are one-dimensional or linear, others like methane have a shape that expands in the three directions of the space–they have a three-dimensional shape. The shape of molecules results from the chemical bonds connecting atoms in a molecule and from the existence of lone pairs of electrons. This chapter covers the analysis of the molecular shape and gains insight into advanced models of the chemical bond. After studying the ideas in this chapter you will be able to draw the connections between the atoms of a molecule and to draft Lewis structures used as graphical representations of the molecular bonds. At the end of the chapter, we will address the idea of molecular polarity, which results from the distribution of charge in a molecule. Polarity will understand the reasons why you use soap to get rid of oil while doing dished.

7.1 Electron-dot structures of atoms

Atoms are made of protons, neutrons, and electrons. Electrons–in particular valence electrons–are responsible for the main chemical properties of an atom. These electrons are tied weakly to the nucleus in comparison with the core electrons and hence they can be exchanged easily with other atoms. Atoms in a molecule, with a few exceptions such as the hydrogen case, will tend to be surrounded by eight electrons so that its electron configuration resembles a noble gas. This is known as the octet rule (the duet rule in the case of hydrogen), and this rule is responsible for the common negative charge of F and the positive charge of Na: F ($[He]2s^2 2p^5$) can easily receive an extra electron producing ionic F^- ($[He]2s^2 2p^6$=[Ne]) and atomic Na ($[Ne]3s^1$) can lose an electron producing ionic Na^+ ($[He]2s^2 2p^6 = [Ne]$). The electron-dot structure of an atom or a molecule is a visual representation of the electronic arrangement in an atom or a molecule.

`Valence electrons` The electrons of an atom can be divided in core electrons and valence electrons. The valence electrons of an atom are involved in chemical bonds as they are less bonded to the nucleus. The number of valence electrons in an atoms is the same as the group number. As an example, hydrogen H belong to the group IA and hence has one valence electrons. Similarly, oxygen O belongs to the group VIA and therefore it has six valence electrons.

Sample Problem 1

Indicate the number of valence electrons for the following atoms: N, O, C and S.

SOLUTION

Nitrogen is in group VA and hence it has five valence electrons ($5e^-$). Oxygen belongs to the group VIA and C belong to IVA, hence they have

STUDY CHECK

Indicate the number of valence electrons for the following atoms: Cl and B.

The octet rule Atoms gain or lose electrons when they combine to form molecules. This electron exchange is the driving force that drives the formation of molecules from single atoms. The octet rule claims that each atom in a molecule is normally surrounded by eight (octet) electrons. There are two important exceptions to this rule as H is only surrounded by two electrons (this is called the duet rule), and B by six.

Electron-dot structure of an atom In order to write the electron-dot structure of an atom, you just need to write down the symbol of the atom surrounded by the number of valence electrons one by one four directions, and if you have more than four electrons then add remaining electrons as pairs. For example, oxygen has six valence electrons and hence, the electron-dot structure would be $\cdot\ddot{\underset{\cdot}{\mathrm{O}}}:$ Similarly, for the case of fluorine the the electron-dot structure would be $:\ddot{\underset{\cdot}{\mathrm{F}}}:$ In the case of an ion, you need to add (if its an anion) or substract (if its a cation) electrons, and for example the electron-dot structure of O^{2-} is $:\ddot{\underset{\cdot}{\mathrm{O}}}:^{2-}$

Sample Problem 2

Write down the electron-dot structure for the following atoms: N, C and Cl^-.

SOLUTION

N has five valence electrons, whereas C has four. Hence the electron-dot for both will be: $\cdot\ddot{\underset{\cdot}{\mathrm{N}}}\cdot$ and $\cdot\dot{\underset{\cdot}{\mathrm{C}}}\cdot$. Cl^- has eight valence electrons, that is seven plus one, and hence its electron-dot structure will be $:\ddot{\underset{\cdot}{\mathrm{Cl}}}:^-$.

STUDY CHECK

Write down the electron-dot structure for N^{3-}

Electron-dot structure of diatomic molecules This section will address how to build up electron-dot structures of diatomic molecules. These would be the most simple electron-dot structures of molecules that you will see. The first step is (a) to set up the atoms in the molecule in the form of a line. After that, (b) you need to count the total number of valence electrons in the molecule by adding the valence electrons of each atom (remember the number of valence electrons corresponds to the group number in the A notation, O is 6 as its group is VIA). Then you (c) compute the pairs of electrons–the total number of valence electrons divided by two; pairs of electrons are represented by lines. In the following (d) you need to start distributing the pairs in the molecule in a very specific way: first connecting the atoms among themselves, and after placing the remaining pairs surrounding the atoms. Overall, each atom should be surrounded by four pairs with the exception of H and B.

Sample Problem 3

Construct the electron-dot structure of HCl.

SOLUTION

1. **Step one:** we first arrange the atoms in the molecule as H Cl.
2. **Step two:** now we count the number of valence electrons: H(1) and Cl(7) that gives a total of eight electrons.
3. **Step three:** let us count the pairs of electrons; we have eight electrons and that is four pairs.
4. **Step four:** now we distribute the pair on each atoms knowing that each atom has to have four pairs with the exception of hydrogen that can only be surrounded by one pair. H:Cl:, using lines instead of pairs (this is not necessary but makes the electron-dot structure look better) we obtain H − Cl|.

STUDY CHECK

Construct the electron-dot structure of HF.

Electron-dot structure of general molecules Now we will address how to build up electron-dot structures of more complex molecules. The first step is (a) to arrange the atoms in the molecule, in the form of a central atom and the remaining atoms around it; the central atom is the one with a lower index in the molecule (e.g. in H_2O is O or in NH_3 is N). After that, (b) you need to count the total number of valence electrons in the molecule, by adding the valence electrons of each atoms. Then you (c) calculate the pairs of electrons–the total number of valence electrons divided by two; pairs of electrons are represented by lines. In the following (d) you need to start distributing the pairs in the molecule in a very specific way (this is the key to building good electron-dot structures): first connecting the surrounding atoms to the central atom, after placing pairs on top of the surrounding atoms and finally by placing the remaining pairs in the central atom. Each atom should be surrounded by four pairs (this is the octet rule) with the exception of H and B as they do not follow the octet rule. When you have the final electron-dot structure, the pairs of electrons (or lines) that connect two atoms are called *bonds*, whereas the pairs not involved in an connection are called *lone pairs*. A very important note is that, at this point, is not that important the atom arrangement (if the molecule looks like a line, a triangle or so) as long as the connectivity (which atom goes in the center and in the surroundings) is correct.

Sample Problem 4

Construct the electron-dot structure of H_2O indicating the number of bonds and lone pairs.

SOLUTION

1. **Step one:** we first arrange the atoms in the molecule as H O H. The central atom is O as oxygen has the lower index in the H_2O molecule–the index for O is one and the index for H is two.
2. **Step two:** now we count the total number of valence electrons, including all atoms: 2xH(1) and O(6) that gives a total of eight electrons.
3. **Step three:** let us count the pairs of electrons; we have eight electrons and that is four pairs.
4. **Step four:** now we distribute the pair on each atoms knowing that each atom has to have four pairs with the exception of hydrogen that can only

be surrounded by one pair. H:Ö:H: and using lines instead of pairs (this is not necessary but makes the electron-dot structure look better) we obtain H−Ō−H . The molecule has two bonds, each one connecting and H to the oxygen atom and two lone pairs located on the oxygen atom.

STUDY CHECK

Construct the electron-dot structure of NH_3 indicating the number of bonds and lone pairs.

Atomic charges in a molecule In order to build up the electron-dot structures of a molecule you needs to count the number of valence electrons. Each atom bring a different number of valence electrons to a molecule. For example, H brings one whereas O brings two. When you arrange the electron pairs in the molecule, each atoms should have the number of electrons that they bring. For example in this electron-dot structure H:C̈l: the hydrogen atom brings one electrons to the molecule, and in the molecule the atom owns one electrons, as in H: one of the dots belong to H and the other belongs to the Cl–the electrons are shared in a covalent bond. In the same way, the Cl atom brings seven electrons and in the molecule it owns seven electrons, and has six electrons here C̈l: plus the one that shared with the hydrogen atom. In another words, the H:C̈l: electron-dot structure is the combination of H· and ·C̈l:. We say that the charges on each atom are zero, as each atom in the molecule owns the same number of electrons that it originally brings.

Sample Problem 5

Indicate the atomic charges on each of the atoms of H−C̄(−H)−H

SOLUTION

The carbon atoms brings four electrons and in the molecule it is surrounded by eight electrons, five of which belongs to it. Hence the charge of C is -1; this means that carbon has one extra electron. Each hydrogen brings one electron and in the molecule each hydrogen has one electron (they share two electrons with C, one for C and one for H).

Multiple bonds Often times you are going to encounter electron-dot structures like :N≡N: in which the atoms are connected by means of a triple bond. You will also encounter double bonds :Ö=Ö: and single bonds. These are called multiple bonds and are formed while constructing electron-dot structures in order to avoid having atoms with atomic charges different than zero. Charged atoms in a molecule are in general not favorable. When you build electron-dot structures and you end up having large atomic charges, you can avoid that by moving electrons from the atom into the bond, hence creating multiple bonds.

Sample Problem 6

Construct the electron-dot structure of O_2.

SOLUTION

1. **Step one:** we first arrange the atoms in the molecule as O O .

2. **Step two:** now we count the total number of valence electrons, including all atoms: 2xO(6) that gives a total of twelve electrons.
3. **Step three:** let us count the pairs of electrons; we have twelve electrons and that is six pairs.
4. **Step four:** now we distribute the pairs, fist connecting the atoms $O-O$ (we have five extra pairs to distribute at this point).
5. **Step five:** we place the remaining pairs on top of the oxygen atoms $|\overline{\underline{O}}-O\rangle$
6. **Step six:** now we calculate the charge on each atom $|\overline{\underline{O}}^{\ominus}-{}^{\oplus}O\rangle$
7. **Step seven:** in order to eliminate the charges we move lone pairs into the bond $\langle O=O\rangle$ Now the charges are zero and this is more important than imposing the octet rule.

STUDY CHECK

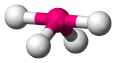
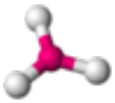
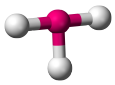
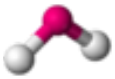
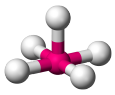
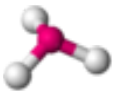
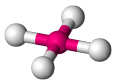
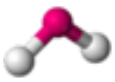
Construct the electron-dot structure of CO_2.

7.2 Molecular shape

Molecules are arrangements of atoms, and these arrangements can be presented in different form. Let us use as an example the H_2O molecule, which contains two hydrogen atoms and one oxygen. Knowing that both hydrogens are connected to oxygen by means of a covalent bond, one can envision several molecular geometries such as $H-\overline{\underline{O}}-H$ or maybe $H\langle\overline{\underline{O}}\rangle H$. The goal of this section is to identify the geometry of a given molecule. In order to do this, the electron-dot structure of the molecule are the key.

ABE Molecular code If the molecule contains two atoms, there is only a possible geometry these two atoms can exhibit, and this is a linear arrangement. For the case of more complex molecules, in order to identify the geometry you need to figure out the ABE code of the molecule. In this code B refers to the number of atoms connected to the central atom in a molecule, and E is the number of lone pairs on the central atom. For example, the electron-dot $H-\overline{\underline{O}}-H$ structure has two bonds with the central atom B_2 and two lone pairs on top of the central atom E_2 and hence the ABE code of the molecule would be AB_2E_2. Another example the ABE code for ammonia $\begin{array}{c}H-\overline{N}-H\\ \quad\;\;|\\ \quad\;\;H\end{array}$ would be AB_3E, as the molecule has three atoms connected to the central nitrogen and and N has a single lone pair. You can find a list of the equivalence between ABE codes and the molecular geometry in Table **??**. In order to predict the geometry of a molecule, once you have the ABE code, Table **??** will give you the geometry. For example, an AB_2 molecule will be linear, whereas an AB_2E_2 is bent. The bond angles are also indicated in the table, and for example a CO_2 molecule, which will be linear will have a $180°$ angle. This means both C-O bonds will form a line.

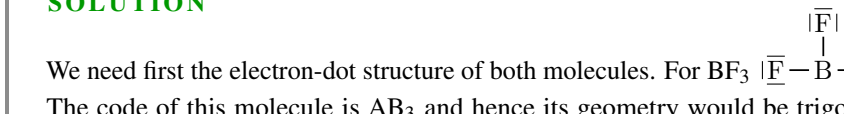
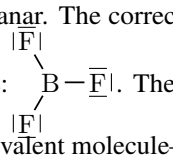
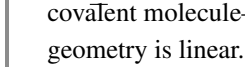
Table 7.1 Molecular geometries

ABE Code	Molecular shape	Bond Angle	3D model	ABE Code	Molecular shape	Bond Angle	3D model
AB_2	Linear	180°		AB_4E	see-saw	180°,120°, 90°	
AB_3	Trigonal Planar	120°		AB_3E_2	T-shaped	90°, 180°	
AB_2E	Bent	120°		AB_2E_3	Linear	180°	
AB_4	Tetrahedral	109°		AB_5E	square pyramidal	90°	
AB_3E	Trigonal pyramidal	109°		AB_4E_2	square planar	90°, 180°	
AB_2E_2	Bent	109°					
AB_5	trigonal bipyramidal	90°, 120°,180°					
AB_6	octahedral	90°, 180°,180°					

Sample Problem 7

Identify the geometry of the following molecules: BF_3 and SO_2.

SOLUTION

We need first the electron-dot structure of both molecules. For BF_3 
The code of this molecule is AB_3 and hence its geometry would be trigonal planar. The correct way to draw the molecule respecting its geometry would be:  . The electron-dot structure for sulfur dioxide–remember this is covalent molecule–is  and its class is AB_2E. Hence the molecular geometry is linear.

STUDY CHECK

Identify and draw the geometry of methane (CH_4).

7.3 Polarity of molecules

This section deals with bond and molecule polarity. A chemical bond will be polar or nonpolar depending on the tendency of the atoms in a bond to attract the electrons the bond. Polar bonds results in the existence of a permanent dipole moment that makes a molecule polar. Polar molecules can interact with polar molecules and mix.

Bond polarity Let us compare two different molecules: H_2 and HCl. We say H_2 is a non-polar molecule. The reason for this is that each atom in the covalent H-H bond equally share the electrons. Differently, HCl is a polar molecule, as H is an electropositive atom and Cl is electronegative. That implies that in the H-Cl covalent bond each atoms shares the electrons in the bond differently. H will be less prone to attract the electrons and Cl would tend to attract the bond electrons more than H. The result would be that the electrons in the bond would belong more to Cl than to H. Another consequence is that the molecule would have a permanent dipole–a permanent charge distribution–result of a uneven charge distribution in the chemical bond. We represent excess of charge as on Cl as $Cl^{\delta-}$ and electron deficiency in H as $H^{\delta+}$. The polarity of the bond is represented as:

$$\overset{+}{H}-\overset{-}{Cl}$$

Polarity of diatomic molecules Molecules can either be polar or non-polar. The polarity of diatomic molecules only depends on the nature of the atoms that forms the molecule. If the atoms in the molecule are the same (e.g. H_2 or O_2), then the molecule would be non-polar. If the atoms are different then the molecule would be polar. Examples are H_2 a nonpolar molecule and HCl or HBr, both polar molecules. You can apply the same concept to a bond inside a molecule. The C-O bond in a CO_2 molecule is a polar covalent bond, as C and O have different electronegativities.

Water (H_2O) is a polar molecule. The ABE type of water of AB_2E_2 and hence its geometry is bent. That means both H-O bonds, which are polar, do not compensate with each other. Hence, the molecule will have a dipole moment and hence will be polar. Methanol (CH_3OH) is a polar molecule as well. The central atom of the molecule (C) is connected to three hydrogens and a OH group. Hence this will be a polar molecule as one of the atoms attached to carbon is different. Both molecules, water and methanol, will mix as they have the same polarity. Methane (CH_4) is a nonpolar molecule, as the four polar C-H bonds compensate each other. Similarly, CCl_4, tetrachloro methene, is another nonpolar molecule, for the same reason. Both molecules, CH_4 and CCl_4 will mix together. As a general rule: molecules with the same polarity (polar-polar or nonpolar-nonpolar) will mix.

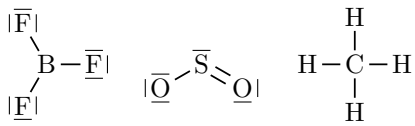
Polarity of larger molecules The polarity of larger molecules would depend on the molecular geometry. Let us analyze the case of CO_2. Each of the C-O bonds on the molecule are polar bonds. However, CO_2 is a linear molecule $\langle O=C=O \rangle$ and the polarity of each C-O bonds compensate so that at the end the molecule is polar. For the H_2O case, again, the H-O bond is polar. However, the molecule is bent and looks just like $H\overset{\overline{O}}{\langle}H$. Both H-O bonds do not compensate as they point in different direction and the directions do not cancel out what makes the water molecule to be a polar molecule.

Sample Problem 8

Identify the polar character (polar/nonpolar) of the following molecules: BF_3, SO_2 and CH_4.

SOLUTION

Let us analyze the geometries of the three molecules:



The bonds on SO_2 do not cancel out, as they do not point in opposite directions. Hence this molecule is polar. On the other hand, the bonds on methane and BF_3 cancel each other out and hence even when the C-H and B-F bonds is polar, these two molecules would be non-polar.

STUDY CHECK

Identify the polar character (polar/nonpolar) of the following molecules: O_2 and NH_3.

Polarity and mixing When you mix two different liquids or even gases, polarity is the key for the mixing process. If the molecules have the same polar character they will be able to mix, whereas they will not mix when the polar character is different. This section will cover several examples of mixing an polarity.

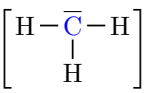
Molecules with the same polarity Water (H_2O) is a polar molecule. The ABE type of water of AB_2E_2 and hence its geometry is bent. That means both H-O bonds, which are polar, do not compensate with each other. Hence, the molecule will have a dipole moment and hence will be polar. Methanol (CH_3OH) is a polar molecule as well. The central atom of the molecule (C) is connected to three hydrogens and a OH group. Hence this will be a polar molecule as one of the atoms attached to carbon is different. Both molecules, water and methanol, will mix as they have the same polarity. Methane (CH_4) is a nonpolar molecule, as the four polar C-H bonds compensate each other. Similarly, CCl_4, tetrachloro methene, is another nonpolar molecule, for the same reason. Both molecules, CH_4 and CCl_4 will mix together. As a general rule: molecules with the same polarity (polar-polar or nonpolar-nonpolar) will mix.

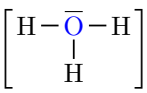
Molecules with different polarity CCl_4 is a nonpolar molecule, and H_2O is a polar molecule. As both have different polar character they will not mix together. If you mix water and CCl_4, tow phases will remain instead of a single mixed liquid phase. As a general rule: molecules with different polarity (polar-nonpolar) will not mix. Another example will be water and oil. Water is polar, and oil is a nonpolar molecule. As a consequence these two molecules will not mix together. Soap has a polar and non-polar part. In order to remove oil from water, soap helps mixing both polar water and nonpolar oil.

CHAPTER 7

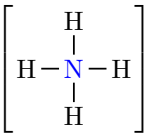
ELECTRON-DOT STRUCTURES OF MOLECULES

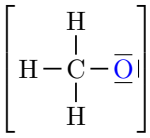
7.1 Indicate the charge of the atom marked blue in the following electron-dot structure:

(a) 

(b) 

7.2 Indicate the charge of the atom marked blue in the following electron-dot structure:

(a) 

(b) 

7.3 Draw the electron-dot structure of: (a) BH_3 (b) CH_4

7.4 Draw the electron-dot structure of: (a) NH_4^+ (b) H_3O^+

7.5 The electron-dot structure of HI is:

(a) $|\underline{H}-\overline{\underline{I}}|$

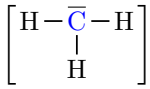
(b) $|\underline{H}-\overline{\underline{I}}$

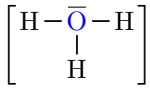
(c) $|\underline{H}-\underline{I}$

(d) $\overline{\underline{H}}-\overline{\underline{I}}$

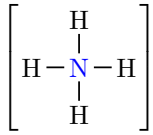
(e) $H-\overline{\underline{I}}|$

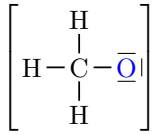
7.6 Indicate the charge of the atom marked blue in the following electron-dot structure:

(a) 

(b) 

7.7 Indicate the charge of the atom marked blue in the following electron-dot structure:

(a) 

(b) 

MOLECULAR SHAPE

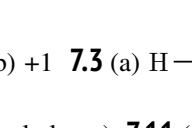
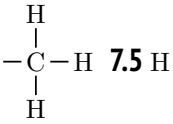
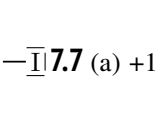
7.8 Identify the molecular shape of the molecules: NH_3 CH_4

7.9 Identify the molecular shape of the molecules: (a) H_2 (b) $BeCl_2$ BF_3

POLARITY

7.10 Indicate the polarity or non-polarity for the following molecules: (a) H_2O (b) HCl (c) H_2

7.11 Indicate the polarity or non-polarity for the following molecules: (a) NH_3 (b) CO_2

Answers **7.1** (a) -2 (b) +1 **7.3** (a)  (b)  **7.5**  **7.7** (a) +1 (b) -1 **7.9** (a) H_2 (Linear) (b) $BeCl_2$ (Linear) BF_3 (Trigonal planar) **7.11** (a) NH_3 (b) CO_2